University of Florida, Department of Mathematics

Analysis PhD Exam, May 2019

Do six problems.

1. Let X, Y be topological spaces. Prove that if $f : X \to Y$ is continuous, then f is Borel measurable.

2. Rigorously determine the following expressions:

(a)
$$\lim_{n\to\infty} \int_0^\infty \frac{\sin(x/n)}{(1+x/n)^n}\, dx$$

(b)
$$\lim_{n\to\infty} \int_0^\infty \frac{n\sin(x/n)}{x(1+x^2)}\, dx$$

(c)
$$\lim_{n\to\infty} \int_0^\infty \frac{n}{n^2+x^2}\, dx$$

3. Prove that if a sequence of functions converges in L^1, then it has a subsequence that converges pointwise almost everywhere.

4. State and prove the Hahn decomposition theorem. For the proof, you may want to consider a sequence of sets such that $\mu(E_n) \to \sup_E \mu(E)$ rapidly, under the assumption that such a supremum is bounded.

5. Let X be a Banach space. Let $Y, Z \subseteq X$ be closed subspaces such that $Y + Z = X$ and $Y \cap Z = \{0\}$. Prove there is $C > 0$ such that for all $y \in Y$ and $z \in Z$,

$$C(\|y\| + \|z\|) \le \|y+z\| \le \|y\| + \|z\|.$$

6. Let X be a Banach space. Show that the natural map from X to X^{**} is injective.

7. State and prove the Cauchy–Schwarz inequality over a complex Hilbert space.

8. Prove or give a counterexample: If f, g are smooth functions on $\mathbb{R}$ such that $f, g \in L^1(\mathbb{R})$, then $f * g$ is a smooth function and $f * g \in L^1(\mathbb{R})$.